



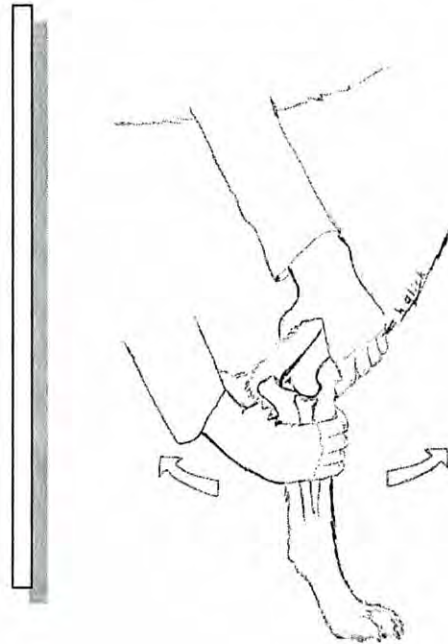
Orthopedic Examination Basics of Sled-Dogs

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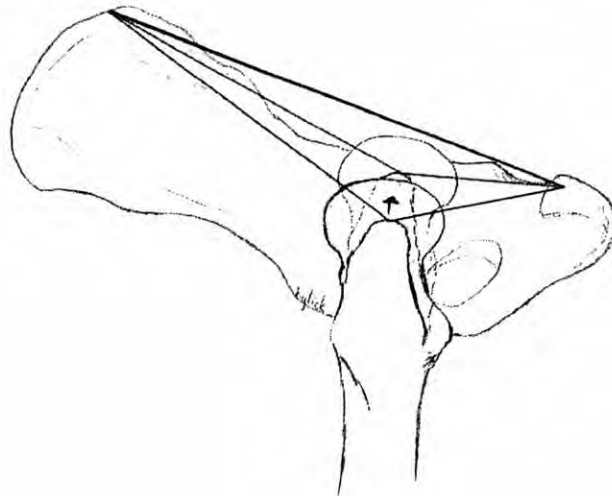
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Orthopedic Examination Basics for the Sled-Dog

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Term glossary for non-veterinarians:

Cranial – Towards the head
Caudal – Towards the tail
Proximal – Towards the body
Distal – Away from the body
Medial – Towards midline
Lateral – Towards the outside
Dorsal – Towards the back, or front aspect of the lower front or hind limb
Palmar – Caudal aspect of the lower front limb
Plantar – Caudal aspect of the lower hind limb
Radiographs – X-rays
Scapula – Shoulder blade
Humerus – Upper arm
Radius/Ulna – Lower arm
Carpus – Wrist
Metacarpus – Bones beneath the wrist
Pelvis – Hip
Femur – Thigh bone
Patella – Knee cap
Stifle – Knee
Tibia – Shin bone
Tibial tuberosity – Crest of the tibia in the front of the bone
Fibula – Small bone next to the tibia
Tarsus – Hock
Metatarsus - Bones below the hock
Flexion – Bending
Extension – Straightening
Crepitus – Grinding
Muscle atrophy - Wasting or decreased amount of muscle
Interdigital webbing - Skin between the toes
Vertebral column - back bone

Lameness may be the result of injury to bone, joints, tendon, muscle, ligaments or neurovascular structures. Therefore lameness may be variable, depending on localization and severity of injury. The orthopedic examination of sled dogs will mostly reveal injuries to the musculoskeletal system. This chapter focuses on techniques to diagnose an orthopedic injury or underlying orthopedic condition. For comparative reasons, a table showing normal joint range of motion values in the dog is provided (Table 1). In rare cases disorders of the nervous system may be the cause of lameness. For that reason, basic steps of a neurological examination are explained at the end of this chapter.

It is imperative to have a basic knowledge of the anatomy of the canine musculoskeletal system when performing an orthopedic examination. This will also help to understand the anatomic terms used in the paragraphs below. Providing the reader with this information is beyond the scope of this chapter. However, there is a tremendous amount of literature available regarding the anatomy of the dog. Canine skeletons can be purchased through various sources.¹ Books on canine anatomy are readily available.^{2,3} If you have the chance to observe a necropsy of a dog, ask the veterinarian to show you the most important muscles, tendons, bones and joints. The more anatomic knowledge you have, the easier it will be to localize and possibly treat the area of discomfort.

The goal of this chapter is to improve your skills when performing an orthopedic examination and to help diagnose orthopedic injuries in sled dogs. Never forget that orthopedic injuries are usually not life-threatening. In contrast, dehydration, poor nutritional status or diseases of the heart, lungs or gastro-intestinal tract are more dangerous to the racing animal than a sore muscle. *Never forget to evaluate these body systems thoroughly prior to the orthopedic examination.*

BASICS

The orthopedic examination includes several steps: 1.) Observation of the gait and obtaining information about the time of onset of lameness. 2.) Performance of a quick overall flexion/extension test. 3.) If necessary, thorough evaluation of the whole leg and all muscles. 4.) Performance of a basic neurologic examination, if indicated.

The gait is best observed when the dog team arrives at a checkpoint. A trail veterinarian should always make a major effort to see the team coming in. It may help to stand at a slight angle to the team when it arrives, and let it pass by. This permits evaluation of the dogs from cranially, laterally, and caudally. Dogs with frontleg lameness usually show a so called "head bob". This is the description of the head going down when the dog places his sound leg on the ground. In contrast, the head goes up when the dog places the injured leg on the ground. In addition, the stride is shortened on the affected side. Just imagine how an injured person walks if hurt on one foot. He or she won't place much weight on the injured leg, but will put more weight on the sound one. *The head falls on the sound leg.*

Dogs with hind leg lameness are a little more difficult to evaluate. The hips of the animal have to be observed. The sore leg prevents full weight bearing, and the hip drops on that side, or the knee may be bent more when compared to the other side. Obviously with very severe injuries, the dog will only toe-touch the ground, or holds up the injured leg.

The history of lameness should be reviewed. The musher should be asked if there are any other dogs he or she may be concerned about. Sometimes dogs are not pulling well or show signs of mild lameness during the run, but perk up when reaching the checkpoint. Some dogs may only show a gait change, not an obvious lameness. For example, loping instead of trotting. If there is a problem, most mushers gladly share their concerns about a dog that may have an orthopedic injury. Occasionally there may be different opinions on the significance of lameness. It needs to be remembered that the musher has spent a very long time with his dogs, and should know their gait very well. On the other hand, mushers should not forget that trail veterinarians are professionals who have learned to examine dogs over years of study, and have gained a great deal of experience in their professional lives. Some have even specialized in orthopedic injuries. Over time, both mushers and veterinarians learn to associate specific gait abnormalities with certain orthopedic injuries, and are able to determine the significance of each gait abnormality.

While the musher takes the booties off, snacks the dogs, and prepares the straw beds, the "Vet-Book" (if available at the race) can be checked for remarks from the last checkpoints. Also evaluate the overall appearance of the dogs, and observe appetite, urination, and the way they lie down. Some dogs may favor one leg, some may tremble or seem otherwise uncomfortable. Prior to starting the examination, it is never wrong to ask the musher if all dogs can easily be examined or if a dog needs to be held while being examined. Better safe than sorry.

At most checkpoints it isn't possible to take a dog out of harness to perform a thorough evaluation as is taught during veterinary school. However, in some cases it may be best to remove the lame dog from the team, and do a complete, "text-book" orthopedic examination. Depending on the situation, the examination can be performed by one person within a few minutes, or may require the help of an assistant, and the complete process may last longer.

As with every time an animal is approached, it is helpful to talk to the dog making it aware of the presence of the veterinarian. Most dogs will be very tired when arriving at a checkpoint and lie down. It may be necessary to prop them up so you can perform your exam. For safety reasons, it is best to straddle the dog. Always try to keep one arm between the dog's head and the examiner's face. Use the left (right) arm to hold the right (left) front leg of the animal, and always keep an eye on the head of the dog. This may decrease the chance of getting bitten. In the vast majority of cases however, sled dogs are used to being handled and the veterinarian does not have to be concerned.

A dog that is not in harness should be restrained appropriately by an assistant. To hold a dog in an appropriate manner on the ground, the assistant should kneel with the knees perpendicular to the back of the animal. Appropriate placement of the hands is important. For example, if the animal is in left lateral recumbency, the assistant's left hand is placed on the left front leg as close to the body as possible. The left elbow of the assistant is placed onto the neck of the animal. The right hand reaches through the hind legs from cranially, and the left hind leg is grasped firmly as proximal as possible to the body of the dog. This will prevent the dog from getting up, and the veterinarian can thoroughly evaluate the animal.

PROTOCOL AND QUICK EXAMINATION

It is imperative to develop a protocol that is followed during every orthopedic examination. This will prevent forgetting a step and decreases the chance of missing a possibly important injury. In general, the right leg should always be compared with the left. This helps to pick up injuries. Being firm during the examination is recommended, as most sled dogs have a high threshold for pain. Some dogs may show only very subtle signs of pain such as facial wincing, withdrawal of the limb or flinching during the exam. However, too much force should not be applied either. Most mushers do not appreciate hearing their dog crying out when being examined. On the other hand, despite gentle handling, some dogs may be severely injured and react very quickly with vocalizing. To prevent misunderstandings between musher and veterinarian it is important to discuss the cause for discomfort on a professional level.

Obviously, if there is evidence of a fracture or another obvious injury, it may not be necessary to perform a complete orthopedic examination. A dog with an orthopedic injury should show one of the following: swelling (soft or firm), pain on flexion, pain on extension, decreased range of motion, crepitus, muscle atrophy, muscle cramping or spasm.⁴ Pain is elicited easier with acute injuries rather than with chronic ones. Dogs that get up after a rest may show lameness that disappears after warming out of the problem. Some dogs may not show pain when they first arrive, but stiffen as the muscles cool down. Therefore, if the musher has seen a problem with a dog, and on initial examination no injury is found, it is recommended to re-examine the dog in a few hours.

As an overall "quick-check" it may be permissible to flex the carpus, elbow, and shoulder at the same time by grasping the dog's metacarpus, and pull it up towards the lateral chest wall. A healthy dog should be able to flex all these joints without any resistance. A similar test can be performed for the hind leg. The metatarsus is grasped and the leg flexed. An injured animal may resist these tests and the examiner should be alerted. The downside with this test is that several areas are evaluated simultaneously. If a pain response is elicited it will be difficult to assign it to a specific part of the leg. It is then important to isolate the area of interest. When performing a more thorough examination, one hand should always be placed above, and one below the joint or area of bone that is examined.

ORTHOPEDIC EXAMINATION

To perform a detailed orthopedic examination, the following steps should be followed:

- 1.) The toes, pads, and skin of the front legs are examined as follows: The foot is picked up and snow and ice is removed. The toes are spread and the interdigital webbing is evaluated. The pads are inspected as they may be injured (Figure 1). The paws are examined for abrasions, discolorations, puncture wounds or swelling. The toe nails are evaluated. Every digit is flexed and extended, and medial and lateral stress are applied. As an experienced trail veterinarian put it once: "The feet of sled dogs are like the tires of race cars. Only minor changes in quality can make huge differences." This should be kept in mind. *Be very thorough during the examination.*

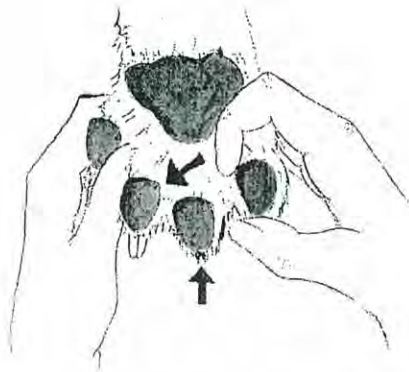


Figure 1: Webbing and pads of the foot are evaluated. Look for lesions on the pads and toe nails (red arrows)

- 2.) Carpal injuries are among the most common problems in the sled dog. The stance and gait of the dog are observed first. It needs to be determined if there is more swelling on one side, or a different angle of the wrist when standing, when compared to the other side. Careful palpation of the dorsal surface of the carpus with the finger tips will help to detect minor swelling (Figure 2). To flex and extend the carpus, the leg is held with one hand proximal the carpus, and the other hand is placed distal to it (Figure 3). Movement of the elbow and shoulder joints should be avoided when performing the evaluation of the wrist. The normal carpus can be flexed to a point where the pad touches the caudal area of the lower leg (Figure 4). The normal extension of the carpus is 10-12 degrees. With the carpus in extension, lateral and medial stability are tested. Instability may be indicative of collateral ligament damage. To do this test, the leg is held above and below the carpus. A certain degree of laxity laterally and medially is normal. It is important to compare the right and left carpus to determine whether the instability that may be felt is normal or pathologic. To detect damage to the accessory carpal bone, pressure is applied to this structure while carefully extending the carpus (Figure 5). A pain response should be followed by taking radiographs. The flexor tendons caudal to the carpus should also be palpated for swelling.



Figure 2: Evaluation for a swollen carpus.



Figure 3: Extension of the carpus.



Figure 4: Flexion of the carpus.



Figure 5: Evaluation for injuries of the accessory carpal bone.

3.) Elbow problems are uncommon in sled dogs. However, this part of the orthopedic examination should not be overlooked. Elbow swelling can sometimes be observed by just looking at the joint and comparing it to the other side, while the dog is standing. This may be difficult in animals with a thick hair coat. Palpation for swelling is done by placing the index finger between the lateral epicondyle of the humerus and the olecranon, and placing the thumb between the medial epicondyle and the olecranon. Both elbows are palpated at the same time. Swelling leads to a bulge between the two bony landmarks when the animal puts weight on the affected leg (Figure 6). The elbow is placed through a range of motion (Figure 7). To evaluate discomfort on the level of the medial coronoid process of the ulna, the carpus can be flexed and rotated internally while holding the elbow slightly flexed. Application of pressure on the medial coronoid process with the thumb of the other hand may elicit a pain response (Figure 8). Joint effusion in association with a pain response to this test is frequently indicative of an injury to the medial coronoid process of the ulna. Collateral ligament injuries are examined next. The carpus and the elbow are flexed and then the paw is rotated internally and externally. With intact medial and lateral collateral ligaments the paw can be rotated 40-50 degrees and 80-90 degrees, respectively. Again, the right and left leg should be compared.



Figure 6: Elbow swelling is evaluated by moving the finger from epicondyle to olecranon and back (blue arrow).

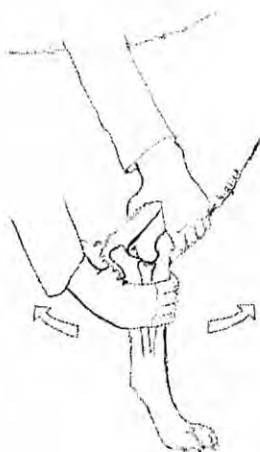


Figure 7: Evaluation for range of motion of the elbow.

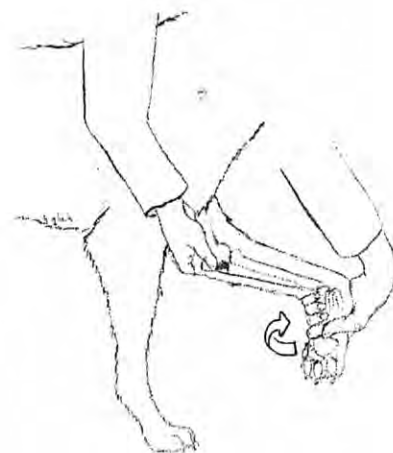


Figure 8: Evaluation for injuries of the medial coronoid process.

4.) Shoulder injuries are among the most common injuries of sled dogs. To examine the canine shoulder, one hand is placed proximal to the elbow on the caudal aspect of the humerus. The other hand is placed over the cranial aspect of the scapula. The shoulder is extended by pushing the humerus forward, and the shoulder joint is stabilized by holding the cranial part of the scapula. Animals that react upon extension of the shoulder may have a joint injury or a strain to the triceps muscle, or other muscles in the caudal area of the shoulder joint. Some biceps injuries will also result in a pain response on extension. The position of the hands is then changed. The shoulder is flexed by pulling the humerus caudally and placing the index- and middle finger of the other hand over the shoulder joint. The bicipital groove is palpated, and some pressure is applied over the biceps tendon. Pain upon flexion of the shoulder can be associated with biceps tendon or biceps muscle injury. The response for these injuries can be exaggerated by deep and thorough palpation of the biceps tendon. With a ruptured biceps tendon, the elbow is straight, compared to an angled elbow joint with intact biceps tendon (Figure 9). Occasionally a pain response when performing this test can result from injuries of the supraspinatus muscle. Injuries to the glenohumeral ligaments or of the joint cartilage may be difficult to diagnose via an orthopedic examination. Some subjective tests to evaluate these structures are described. However, more sophisticated diagnostics such as arthroscopy, computed tomography (CT), or magnetic resonance imaging (MRI) done at a specialty center will likely provide more objective data to diagnose such problems:

Figure 9: Biceps tendon rupture results in a straight elbow when flexing the shoulder joint (red), compared to normal (green).



- 5.) The toes on the hind legs are evaluated as described for the front legs.
- 6.) Tarsal (hock) injuries are not commonly observed in sled dogs. The evaluation of this joint follows for the most part the description given for the carpus (see above). Swelling of the tarsal joint is best felt by palpating the area just proximal to and on each side of the calcaneus (Figure 10). The collateral ligaments can be evaluated with the hock held in extension. The joint is opened laterally and medially.
- 7.) The Achilles tendon consists of five tendons (tendon of the gastrocnemius muscle, common tendon of the biceps femoris, semitendinosus, and gracilis muscles, and the tendon of the superficial digital flexor muscle). A partial Achilles tendon rupture (only the superficial digital flexor tendon is intact) usually results in a typical stance of the animal, with the tarsus hyperflexed (dropped) in comparison to the non-injured leg, and the toes in a flexed "crablike" position. Complete Achilles tendon rupture results in a plantigrade stance, or extreme hyperflexion, of the hind leg (the dog seems to walk on the hock instead of on the toes). Achilles tendon injury or strain will lead to swelling of the tendon. To detect swelling, palpate the gastrocnemius muscles and Achilles tendons on each side simultaneously (Figure 10). The tendon should also be palpated thoroughly during flexion and extension of the hock. Shortly after injury, it may be possible to detect a gap between the calcaneus and the avulsed end of the tendon several centimeters above the calcaneus. Over time this gap will be filled with fibrous scar tissue, possibly giving the examiner the false impression of examining an intact tendon. Radiographs (x-rays) or ultrasound of the tendon may help to support the diagnosis.